

Homework (Jalapeno)

- Q1.** In ancient Egypt, a pharaoh wants to build a rectangular pyramid with a base perimeter of P and a height of H . The pharaoh has the following constraints: $2x + 3y \leq 24$, $x + 2y \leq 16$, $x \geq 0$, and $y \geq 0$. Which of the following is a feasible solution?
- (A) $x = 4, y = 2$
 (B) $x = 6, y = 4$
 (C) $x = 8, y = 0$
 (D) $x = 0, y = 8$
 (E) $x = 2, y = 6$
- Q2.** The Great Wall of China has a series of watchtowers that are spaced at equal intervals. If the distance between two consecutive towers is d , and the total length of the wall is L , which inequality represents the maximum number of towers that can be built?
- (A) $d \geq L$
 (A) $d \leq L$
 (A) $n \geq L$
 (A) $n \leq L$
 (A) $n = L$
- Q3.** A historian is studying the population growth of ancient Mesopotamia. The population P is modeled by the inequality $P \leq 2t + 10$, where t is the time in years. Which of the following values of t satisfies the inequality when $P = 25$?
- (A) $t = 5$
 (B) $t = 7$
 (C) $t = 10$
 (D) $t = 12$
 (E) $t = 15$
- Q4.** The Rosetta Stone is an ancient artifact that has a rectangular shape with a length l and a width w . The stone has the following constraints: $l + 2w \leq 20$, $2l + w \leq 25$, $l \geq 0$, and $w \geq 0$. Which of the following is a feasible solution?
- (A) $l = 8, w = 4$
 (B) $l = 10, w = 5$
 (C) $l = 12, w = 6$
 (D) $l = 15, w = 7$
 (E) $l = 20, w = 0$
- Q5.** The construction of the Colosseum in ancient Rome required a large amount of stone. The amount of stone S is modeled by the inequality $S \geq 3x + 2y$, where x is the number of columns and y is the number of arches. Which of the following values of x and y satisfies the inequality when $S = 50$?
- (A) $x = 10, y = 5$
 (B) $x = 12, y = 6$
 (C) $x = 15, y = 7$
 (D) $x = 18, y = 8$
 (E) $x = 20, y = 10$
- Q6.** The Terracotta Army in ancient China has a series of soldiers that are arranged in a rectangular formation with m rows and n columns. The total number of soldiers T is modeled by the inequality $T \leq mn$. Which of the following values of m and n satisfies the inequality when $T = 100$?
- (A) $m = 5, n = 20$
 (B) $m = 10, n = 10$
 (C) $m = 15, n = 6$
 (D) $m = 20, n = 5$
 (E) $m = 25, n = 4$
- Q7.** The Hanging Gardens of Babylon are a series of gardens that are supported by a system of arches. The height h of the gardens is modeled by the inequality $h \leq 2x + 3y$, where x is the number of arches and y is the number of pillars. Which of the following values of x and y satisfies the inequality when $h = 15$?
- (A) $x = 3, y = 2$
 (B) $x = 4, y = 3$
 (C) $x = 5, y = 4$
 (D) $x = 6, y = 5$
 (E) $x = 7, y = 6$
- Q8.** The Great Pyramid of Giza has a square base with a side length s . The height h of the pyramid is modeled by the inequality $h \leq s + 10$. Which of the following values of s satisfies the inequality when $h = 20$?
- (A) $s = 5$
 (B) $s = 10$
 (C) $s = 15$
 (D) $s = 20$
 (E) $s = 25$

Open Ended Questions (FRQ)

- Q9.** The ancient city of Pompeii has a rectangular wall with a length l and a width w . The wall has the following constraints: $l + 2w \leq 30$, $2l + w \leq 35$, $l \geq 0$, and $w \geq 0$. Find the maximum area of the wall and the corresponding values of l and w . Show your work and explain your answer.
- Q10.** The ancient temple of Angkor Wat has a series of towers that are spaced at equal intervals. The distance between two consecutive towers is d , and the total length of the temple is L . The temple has the following constraints: $d \geq 5$, $L \leq 100$, and $n \geq 2$, where n is the number of towers. Find the maximum number of towers that can be built and the corresponding value of d . Show your work and explain your answer.

Chef's Tips

- Tip 1: Make sure to read each question carefully and understand the constraints.
- Tip 2: Use graphical methods to visualize the solution regions.
- Tip 3: Check your work and explain your answers clearly.